```
  Performing 'strings' command checks
    Checking 'strings' command                          [ OK ]

  Performing 'shared libraries' checks
    Checking for preloading variables                   [ None found ]
...
```

This check will take some time to run but we have bypassed the user interaction prompts with the `-sk` argument. Afterwards, you will receive a system check summary detailing what was found.

Answer the questions below

Run *chkrootkit* on the affected system. What is the full path of the `.sh` file that was detected?

/var/tmp/findme.sh

✓ Correct Answer

Run *rkhunter* on the affected system. What is the result of the `(UID 0) accounts` check?

Warning

✓ Correct Answer

Task 8 Conclusion

How likely are you to recommend this room to others?

1 2 3 4 5 6 7 8 9 10

After making an SUID copy of `/bin/bash`, the attacker elevated to root by running `/var/tmp/bash -p`. We can further verify the `bash` binary by performing an integrity check on the original:

Integrity Checking the Suspicious SUID Binary

```
investigator@MACHINE_IP:~$ md5sum /var/tmp/bash
7063c393************d3b340f1ad2c  /var/tmp/bash
investigator@MACHINE_IP:~$ md5sum /bin/bash
7063c393************d3b340f1ad2c  /bin/bash
```

The output above shows that the two binaries are identical, further enhancing our understanding of the attacker's actions to escalate to root.

Answer the questions below

Run the `debsums` utility on the compromised host to check only configuration files. Which file came back as altered?

/etc/sudoers

✓ Correct Answer

What is the `md5sum` of the binary that the attacker created to escalate privileges to root?

7063c3930affe123baecd3b340f1ad2c

✓ Correct Answer

Task 7 Rootkits

Task 8 Conclusion

How likely are you to recommend this room to others?

```
Birth: -
```

If we look back to the output of the `ls -al` command, we can identify the permission misconfiguration that made this possible:

Viewing the Permissions of the authorized_keys File

```
investigator@MACHINE_IP:~$ ls -al /home/jane/.ssh/authorized_keys
-rw-rw-rw- 1 jane jane 1136 Feb 13 00:34 /home/jane/.ssh/authorized_keys
```

As identified by the third `rw` permissions, this file is world-writable, which should never be the case for sensitive files. Consequently, by exploiting this misconfiguration, the attacker gained unauthorised SSH access to the system as if they were Jane.

Answer the questions below

View Jane's `.bash_history` file. What flag do you see in the output?

THM{f38279ab9c6af1215815e5f7bbad891b}

✓ Correct Answer

What is the hidden flag in Bob's home directory?

THM{6ed90e00e4fb7945bead8cd59e9fcd7f}

✓ Correct Answer

Run the `stat` command on Jane's `authorized_keys` file. What is the full timestamp of the most recent modification?

2024-02-13 00:34:16.005897449 +0000

✓ Correct Answer

Task 6 Binaries and Executables

More specifically, this line specifies:

- **richard** is the username being granted sudo privileges.
- **ALL** indicates that the privilege applies to all hosts.
- **(ALL)** specifies that the user can run the command as any user.
- `/sbin/ifconfig` is the path to the specific binary, in this case, the ifconfig utility.

With this configuration, Richard can execute `ifconfig` with elevated sudo privileges to manage network interfaces as necessary.

Answer the questions below

Investigate the user accounts on the system. What is the name of the backdoor account that the attacker created?

b4ckd00r3d ✓ Correct Answer

What is the name of the group with the group ID of **46**?

plugdev ✓ Correct Answer

View the `/etc/sudoers` file on the compromised system. What is the full path of the binary that Jane can run as sudo?

/usr/bin/pstree ✓ Correct Answer

Task 5 User Directories and Files

Task 6 Binaries and Executables

```
Device: ca01h/51715d    Inode: 526643      Links: 1
Access: (0755/-rwxr-xr-x)  Uid: (   33/www-data)   Gid: (   33/www-data)
Access: 2024-02-13 02:31:29.256000000 +0000
Modify: 2024-02-13 00:26:28.000000000 +0000
Change: 2024-02-13 00:34:50.679215113 +0000
 Birth: -
```

Answer the questions below

To practice your skills with the `find` command, locate all the files that the user **bob** created in the past 1 minute. Once found, review its contents. What is the flag you receive?

THM{0b1313afd2136ca0faafb2daa2b430f3}

✓ Correct Answer

Extract the metadata from the `reverse.elf` file. What is the file's MIME type?

application/octet-stream

✓ Correct Answer

Run the `stat` command against the `/etc/hosts` file on the compromised web server. What is the full **Modify Timestamp (mtime)** value?

2020-10-26 21:10:44.000000000 +0000

✓ Correct Answer

Task 4 Users and Groups

Task 5 User Directories and Files

Task 6 Binaries and Executables

Task 7 Rootkits

We can modify our `PATH` and `LD_LIBRARY_PATH` (shared libraries) environment variables to use these trusted binaries:

Modifying Environment Variables to Include Trusted Paths

```
investigator@MACHINE_IP:~$ export PATH=/mnt/usb/bin:/mnt/usb/sbin
investigator@MACHINE_IP:~$ export LD_LIBRARY_PATH=/mnt/usb/lib:/mnt/usb/lib64
```

Answer the questions below

After updating the `PATH` and `LD_LIBRARY_PATH` environment variables, run the command `check-env`. What is the flag that is returned in the output?

THM{5514ec4f1ce82f63867806d3cd95dbd8}

✓ Correct Answer

Task 3 Files, Permissions, and Timestamps

Task 4 Users and Groups

Task 5 User Directories and Files

Task 6 Binaries and Executables

Task 7 Rootkits

Task 8 Conclusion